

Objective

Electronics Post graduate with 17 Years of hands-on experience in Hardware design, debugging, testing, System Integration test bench design (HIL) & manufacturing, seeking a challenging position where I can contribute to developing innovative solutions and gain expertise in an embedded domain.

Professional Summary (Total 17 years of Experience)

1. Offshore Professional Summary:

- ❖ In the field of Embedded System design, Automotive (HIL Testing) and Microcontroller based IOT application Testing, Robots for Industrial purpose or can be used in automotive for automation testing
- ❖ Hands-on Programming experience on Microcontrollers like 8, 16 & 32 Bit using Embedded C & Assembly (for 8051 and Pic)
- ❖ Experience on Microcontroller programming tools like Keil, MPLAB, Atmel Studio, STM CUBEIDE & IAR workbench
- ❖ Good hands-on experience in Analog & Digital Circuit Design
- ❖ Good hands-on experience in PCB designing (tool used: Protel (From Altium))
- ❖ Good to design a new Embedded System based on Microcontroller, Sensor (Input) & Actuators (Output)
- ❖ Capable of Circuit Analyzing, Testing & Circuit Modification as per the requirement.
- ❖ Good to design the Test bench setups for system integration testing activity like HIL

2. Onsite Professional Summary:

- ❖ Participated in the Technical Discussion during Business meet at Aliaro Head Office, Sweden
- ❖ Worked as a Developer & Tester for eCall in Continental, Singapore
- ❖ Worked as a Project coordinator for HIL Testing in TAKATA, Germany

Technical Skills

- ❖ Automotive (ICE & EV) System Integration Testing & Test system design expert (HIL system design & making a hardware based Setup) using Vector VT System (ICE & EV), dSPACE HIL system, Key-sight Cell Simulator for EV Cell Testing
- ❖ Expert in Wire harness design, Test Bench design and setup, Component list preparation for EV & ICE related Test bench and HIL
- ❖ Worked on 3D printing technology for developing the prototypes using an Embedded System based on Microcontroller and can design the 3D models for test bench and for HIL as per EV requirement.
- ❖ Microcontroller based Embedded System Design only on Windows platform.
- ❖ Expert in interfacing of Memory (EEPROM), 3*3/4*4 Keypad, GSM, RFID, RF, Ultrasonic, LCD & 7 Segment, Bluetooth, Wi-Fi, Zig-Bee and all kinds of Sensor and Actuator with Micro-controller.
- ❖ Expert in Low Level Device Driver and firmware implementation based on Micro-controller like RS232, UART, I2C, SPI, 1-wire protocol.
- ❖ Ability to write the S/W for Mixed Signals, Timers, Interrupts and PWM based on Micro-controller.
- ❖ Ability to Design a Prototype PCB & Developing the same.

Educational Qualification:

- ❖ Certification in Real Time Embedded System from Cranes Varsity, Bangalore, India (2010-2011)
- ❖ M.Tech (Sensor System Technology) from Vellore Institute of Technology, Vellore, India (2007-09)
- ❖ B.E. (E&C) from Institute of Technology and Management, Gurugram, India (1998-2002)

Work Experience:

- 1. Organization: In-house (Self Employed (Jan2025 to Till date))**
Position: - System Architect

Responsibilities: -

- ❖ My role involves a range of activities such as planning, organizing, coordinating, and controlling all the resources required to produce a company's products or service to ensure that business operations are efficient, effective, and capable of meeting customer demands.
- ❖ Gather the requirement from the market and implement in own ideas (Embedded System Design).

Project 1:

- ❖ **Developed a Microcontroller based an embedded system** interfaced with Relays, PWM (for DC Motor driving activity), I2C, SPI, DAC, UART, LIN & CAN to simplify the testing activity of an embedded system (like it can be used in Automotive, Medical, TV industries, Mobile industries etc etc wherever an Electronic system involves for **system integration** testing activity).
- ❖ For Testing activity of own system I have used DMM & Oscilloscope for functionality testing.
- ❖ Designing the PCB (Using Protel Software) for prototype and then do the assembly by own.

Project 2:

- ❖ **Designed a Robot to test the one of the major application of an automotive system using a** low cost controller, geared DC motor and RF module.
- ❖ For Testing activity of own system I have used DMM & Oscilloscope for functionality testing.
- ❖ Designing the PCB (Using Protel Software) for prototype and then do the assembly by own.

2. Organization: Aliaro Technologies, India (Nov 2023 to Dec 2024)

Position: - Project Manager (HIL Design & Manufacturing Team)

Responsibilities: -

- ❖ My role involves a range of activities such as planning, organizing, coordinating, and controlling all the resources required to produce a company's products or service to ensure that business operations are efficient, effective, and capable of meeting customer demands.
- ❖ Involvement of myself in technical discussion with the sales team to understand the client requirements and discussed about the resource management and budget allocation for the project.
- ❖ Involvement of me with the technical team to prepare the Demo Setup for the customers based on their demands.
- ❖ Monitoring HIL project execution like HIL Design, Hardware procurement like Power supply, HIL rack, Wires, connectors, testing the entire HIL, Quality inspection after building the HIL.
- ❖ Creates the JIRA tickets for the progress and problems.
- ❖ Till Dec 2024 from March 2024, I have delivered 4 HIL Projects which includes 19 HIL systems to one of Indian client of Aliaro.

3. Organization: Vector Informatika India Pvt Ltd, India (Nov 2022 to Oct 2023)

Position: - Senior Business development Specialist

Responsibilities: -

- ❖ Understand the use case of the client's requirement and based on that purpose the Vector solutions Like Canoe pro/run/pex, VN xxxx tools for network interface etc.
- ❖ Schedule the Demo at customer place based on their demand. Visiting the customer place for the discussion and to see their setup if the customer has the setup for which they required the solutions from the vector.
- ❖ Dealing with internal team to provide the quick solution to the customer.

4. Organization: OLA Electric, India (Oct 2021 to Oct 2022)

Position: - Technical Consultant /Sr. Manager

Responsibilities: -

- ❖ Functional Testing: HIL setup design for BMS, VCU, MCU & Charger
- ❖ Budget making for HIL Test bench design for entire vehicle testing.
- ❖ Understanding the In-House Project requirements and based on that prepare the list of the components required for Test bench and HIL setup design for functional test case execution.
- ❖ Wire harness Design for BMS, VCU, MCU & Charger for Integrated HIL Setup
- ❖ Designed the Test benches for BMS, VCU & MCU testing activity which includes Dummy Cell simulator, Temperature Simulator & SOC Indicator using passive components likes resistors, Pot's & LED's & Actuators and Sensors

- ❖ Giving the training to the people who are fresh in this EV domain based on Microcontroller and Hardware design.
- ❖ Load Box Design for VCU Software testing activity including the actual loads which are going to be used in actual vehicle.
- ❖ Design the wire harness for VCU & BMS HIL using Key-sight's Cell Simulator and Vector VT Systems

5. Organization: Wipro Ltd, India (June 2018 to Sept 2021)

Position: - Technical Lead

➤ **Client 4: Electrification Practice, Wipro (Feb 2020 to Aug 2021)**

Project: - HIL Testing & Test Bench Design (In house project)

Responsibilities: -

- ❖ Handled the Project of OLA Electric (TC) under the Electrification Practice department for BMS, VCU & MCU testing activity at client place.
- ❖ Understanding the client's requirement and based on that prepare the list of the components required for Test bench and HIL setup design for functional test case execution.
- ❖ Prepared the Wire harness for BMS.
- ❖ Designed the Test bench for BMS testing activity.
- ❖ Giving the training to client's people who are fresh in this EV domain based on Microcontroller and Hardware design.
- ❖ Wire harness Design for VCU & BMS HIL (Keysight's Cell Simulator and Vector VT Systems at client place i.e. OLA Electric)
- ❖ Prepare & fabricate the PCB's for BMS and VCU Test bench.
- ❖ Help the Client's testing team in writing the CPAL scripting for BMS testing activity.
- ❖ Setup a meeting with the suppliers for EV Test Bench Design
- ❖ Preparing the Documents based on the requirement.
- ❖ Giving the training to others (Who are not from Electronics background) on Microcontroller and Hardware design.
- ❖ Coordinating the team members who are working at Client place.

➤ **Client 3: FORD Automotive, USA (Feb 2019 to Jan 2020)**

Project: - Infotainment

Responsibilities: -

- ❖ Handling a team of 5 -10 Tester's (based on the work load)
- ❖ Fixing the issues which was coming during System Testing (Bench side)
- ❖ Cross check the result of Unit & Manual testing on fixed issue done by the team.
- ❖ After successful unit testing done by team, review & approve the code.
- ❖ Prepare WSR report weekly basis and send to RM for review.
- ❖ Create ticket in JIRA tool for any issues like Product Defects

➤ **Client 2 : Continental Automotive, Singapore (Nov 2018 to Jan 2019)**

Project: - Ecall (Emergency Call)

Responsibilities: -

- ❖ Worked on SOC side (GSM Module) to execute the functional test cases.
- ❖ Fixing the issues which was coming during System Testing
- ❖ Perform Unit & Manual Testing on fixed issues.
- ❖ Create ticket in JIRA tool for any issues.

➤ **Client 1 : Continental Automotive, USA (June 2018 to Oct 2018)**

Project: - Telematics Platform

Responsibilities: -

- ❖ Worked on VUC side (On Vehicle Microcontroller)
- ❖ Fixing the issues which was coming during System Testing
- ❖ Perform Unit testing on fixed issue.
- ❖ Prepare ConUps for a feature which is going to add in current platform.
- ❖ Create ticket in JIRA tool for any issues.

6. Organization: Brisa Technologies Pvt Ltd, India (June 2017 to Feb 2018)

Position: - Technical Architect

➤ **Client:-** Cypress Semiconductor Corporation, India (June 2017 to Feb 2018)

Project 1: - BLE (Version 42 & 62) Hardware & Software Validation

Project 2: - DFT testing for Cypress kits (@component level) for Cypress 149, 145 & 62-BLE Kit's

Responsibilities: -

- ❖ Writing & executing the test cases for BLE 42 & 62 Pioneer Kit
- ❖ Validating the Software on BLE 42 & 62 Pioneer Kit
- ❖ Creating CDT in Cypress tool after finding the issue either in Hardware validation or Software
- ❖ Involved into the discussion with the Hardware and Software Developer to fix the issue
- ❖ Prepares the Report for Hardware or Software validation & submit into Cypress tool for Approval

7. Organization: Tech Mahindra, Bangalore (September 2013 to March 2017)

Position: - Technical Lead

➤ **Client 3:-** YAZAKI, Romania (July 2016 – Jan 2017)

Project: - Instrument Panel Cluster (IPC)

Responsibilities: -

- ❖ Cluster requirement analysis and development activities for system test and HIL test
- ❖ Writing test specifications and executing integration test cases
- ❖ Defect logging and debugging the test cases.
- ❖ Simulation and HIL testing of IPC (Cluster) using Canoe and Test Automation Editor (TAE) like Robot Framework

➤ **Client 2:-** TAKATA, Germany (January 2016 – June 2016)

Project: - Motorized Seat Belt (MSB)

Responsibilities: -

- ❖ Developing the Real-time scenario using Controllers for injecting the faults in ECU
- ❖ Writing & executing the test cases for MSB in Canoe tool
- ❖ Simulation and HIL testing of Motorized seat belt using VT system and Test Automation Editor (TAE)
- ❖ Conducted technical training sessions for a diverse team of engineers to enhance knowledge sharing within the team.

➤ **Client 1:-** TAKATA, Germany (September 2013 – Dec 2015)

Project: - Active Front Steering System (ASTU)

Responsibilities: -

- ❖ Developing the Real-time scenario using Controllers for injecting the faults in ECU
- ❖ Writing & executing the test cases for ASTU in Canoe tool
- ❖ Simulation and HIL testing of Active Front Steering System using VT system and Test Automation Editor (TAE)
- ❖ Conducted technical training sessions for a diverse team of engineers to enhance knowledge sharing within the team.
- ❖ Designed a Hardware Automation Circuit (Relay Circuit for High Amp Testing) to protect the VT system and programming without changing any connection.

8. Organization: Cranes Software International Ltd, Bangalore (May 2011 to June 2013)

Position: - Software Engg.

➤ **Client:** - Caravel (May 2011 to June 2013)

Project: - In House Projects (Cranes Software International Ltd.)

Responsibilities: -

- ❖ Working on various project initiatives involving Micro-controllers
- ❖ Conducting corporate training for associates on Micro-controllers

9. Organization: JIET College, Jodhpur (Oct 2009 to April 2010)

Position: - Lecturer in Electronics & Communication department

Subjects:- Microcontroller 8051 & LIC (Linear Integrated Circuits)

Project Coordinator:

- ❖ **Project 1:** Head Light movement of Vehicle through Steering:
The device was designed to save human life on curve roads when headlights of the four wheelers do

not adjust with the steering wheel leading to some blind spot and accident. Project is to develop the mechanism to synchronize movement of headlights with steering wheel.

- ❖ **Project 2:** Bridge Monitoring System Using Fiber Optical Cable as a Sensor:
Project is undertaken to measure the force at the critical point of the Bridge Like Center point using Fiber Optical cable as a Sensor to avoid the accidents on the Bridge.

10. Organization: [Garg Electronics, Chandigarh \(April 2005 to April 2007\)](#)

Position: - Printed Circuit Board & Circuit Designer

- **Client:** - Videocon, BeeTel (Bharti Telecom Ltd), Delphi, Wipro, Osram

Responsibilities: -

- ❖ Understanding the Requirements of the client
- ❖ PCB Designing.
- ❖ PCB Layout modification as per the Customer requirement.

In House Project Summary:

- **Home/Office Automation using RF & IR**
Device was designed as a prototype to switch on/off our home/office's electrical appliance using IR Sensors placed at the doorstep & will send the signal to remotely placed control panel (The Switch Board).
- **Intelligent Traffic Control System Based on IR (Prototype)**
The device was designed to count the number of vehicles & based on this count the system will turn on the green light for that traffic lane & will also provide the time delay to the next lane to turn on the green light from red light for that same time.
- **Industrial Automation using I2C Protocol**
Device was designed to measure the temperature of a furnace & trigger the alarm when it crosses the limit, switch on the water pump in case of fire & open the ventilator gates to exhaust the gases.
- **Remotely Controlled Fire Fighting Robot**
Device designed to help fire fighters in dangerous situations. The device not only acts as fire fighter but also sends the live body counts and temperature data outside for further action.
- **Designed a GSM based switching system for all kinds of electrical appliances**
The device was designed as a prototype for switching on/off for our home/office's electrical appliance from a distance using GSM Technology.

Other Projects as Freelancer:

- **Client:** [Emergers Technologies Pvt, Ltd, Bangalore](#)
Project 1: Fully Automated Water Level Controller using 8051 controller
The device was designed to switch on and off the water pump (motor) automatically after getting the signals from sensors placed in Over Head Water Tank as well as Sump. Also, it makes sound for dry run to protect the motor to burn out.

Project 2: Water purifier motor controller w.r.t time using 8051 controller
The device was designed to switch off the purifier motor after a certain time set by the user. When a user switch on the circuit, motor will be turned off and then User has set the time in the system to ON the motor and once it reaches to 00, motor will turn off and buzzer will ON indicate the user that Motor is OFF.